

A K_8 COUNTEREXAMPLE TO CONJECTURE 3.9 OF BARÁT AND NAGY

ABSTRACT. Barát and Nagy conjectured that every proper edge-coloring of K_{2n} by at least $2n - 1$ colors contains a rainbow 2-regular subgraph on $2n - 1$ or $2n$ vertices. We show that the conjecture fails for $n = 4$. The coloring is the one-factorization represented by matrix B in work of Akbari, Etesami, Mahini, and Mahmoody, who reported that it has no rainbow 7-cycle. We identify this earlier coloring as a counterexample and give a self-contained algebraic proof that it has no rainbow 2-regular subgraph on seven vertices.

A subgraph is *rainbow* if its edges have pairwise distinct colors. We use “2-factor on m vertices” for a 2-regular subgraph with m vertices, as in [2]. Conjecture 3.9 of Barát and Nagy states that every proper edge-coloring of K_{2n} by at least $2n - 1$ colors contains a rainbow 2-factor on $2n - 1$ or $2n$ vertices.¹ Since $\chi'(K_{2n}) = 2n - 1$, a proper $(2n - 1)$ -edge-coloring is a one-factorization, so a one-factorization counterexample refutes the “at least” form as well. We disprove the conjecture at $n = 4$.

Theorem 1. *There is a proper 7-edge-coloring of K_8 with no rainbow 2-regular subgraph on seven or eight vertices. Consequently, Conjecture 3.9 of Barát and Nagy is false.*

Proof. Let

$$\Gamma = \mathbb{F}_2^2 = \{0, a, b, a + b\},$$

and let $\rho: \Gamma \rightarrow \Gamma$ be the linear automorphism defined by

$$\rho(a) = a + b, \quad \rho(b) = a.$$

Then $\rho(a + b) = b$, so ρ cyclically permutes the three nonzero elements of Γ and has no nonzero fixed point.

Take $V(K_8) = \Gamma \times \{0, 1\}$. For $d \in \Gamma \setminus \{0\}$ assign the internal edge $\{(x, i), (x + d, i)\}$ the color I_d , and assign the cross edge $\{(x, 0), (y, 1)\}$ the color

$$C_{\rho(x)+y}.$$

Each I_d is a perfect matching. Each C_s is a perfect matching because $x \mapsto s + \rho(x)$ is a bijection of Γ . Thus the three internal colors and four cross colors form a one-factorization of K_8 .

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¹The statement is Conjecture 3.9 of the published version [2]; it appears as Conjecture 3.8 of the preprint [3].

Suppose that F is a rainbow 2-regular subgraph on seven vertices. It has seven edges and hence uses every color exactly once. In particular, it has four cross edges and three internal edges. Put

$$r_i = |V(F) \cap (\Gamma \times \{i\})|, \quad m_i = |E(F[\Gamma \times \{i\}])|.$$

Degree sums in the two parts give

$$2r_i = 2m_i + 4.$$

Since $r_0 + r_1 = 7$, one part contains three vertices and one internal edge, while the other contains four vertices and two internal edges. Let I_d be the color of the single internal edge and I_e, I_f the colors of the other two. The three internal colors are all used, so

$$e + f = d.$$

Write the cross edges, with incidence labels repeated when necessary, as

$$\{(x_j, 0), (y_j, 1)\}, \quad 1 \leq j \leq 4.$$

In each part, the sum of the labels over all edge incidences is zero, since every vertex of F has degree two and Γ has characteristic two. An internal edge of color I_t has endpoint sum t . The internal-edge sum is therefore d in both parts: it is d on one side and $e + f = d$ on the other. Hence

$$\sum_{j=1}^4 x_j = d, \quad \sum_{j=1}^4 y_j = d.$$

All four cross colors occur once, and the sum of the elements of Γ is zero. Therefore

$$0 = \sum_{s \in \Gamma} s = \sum_{j=1}^4 (\rho(x_j) + y_j) = \rho(d) + d,$$

contradicting the absence of a nonzero fixed point of ρ . Thus no rainbow 2-regular subgraph on seven vertices exists.

A 2-regular subgraph on all eight vertices has eight edges and cannot be rainbow in a coloring with only seven colors. $\square$

Prior art and scope. Under the vertex order

$$(0, 0), (a, 0), (b, 0), (a + b, 0), (0, 1), (a, 1), (b, 1), (a + b, 1)$$

and the color order

$$I_a, I_b, I_{a+b}, C_0, C_a, C_b, C_{a+b},$$

this is exactly matrix B in [1]. Akbari, Etesami, Mahini, and Mahmoodi reported, based on a computer search, that the coloring has no rainbow 7-cycle. The one-factorization and that property are prior. The proof above excludes both possible 2-regular graphs on seven vertices, C_7 and $C_3 \cup C_4$, and records the resulting counterexample to Conjecture 3.9. The analogous statement restricted to $n \geq 5$ is not addressed here.

REFERENCES

- [1] S. Akbari, O. Etesami, H. Mahini, and M. Mahmoody, *On rainbow cycles in edge colored complete graphs*, Australas. J. Combin. **37** (2007), 33–42.
- [2] J. Barát and Z. L. Nagy, *Transversals in generalized Latin squares*, Ars Math. Contemp. **16** (2019), no. 1, 39–47, <https://doi.org/10.26493/1855-3974.1316.2d2>.
- [3] J. Barát and Z. L. Nagy, *Transversals in generalized Latin squares*, preprint, arXiv:1701.08220v1 (2017), <https://arxiv.org/abs/1701.08220v1>.